

SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 06:30 UTC

SEC P vs NP — daily log 2026-04-28

2 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) $\times$ Boolean formula depth lower bounds for KRW-style block compositions $f \circ g$, the central complexity-theoretic object behind $P \not\subseteq NC^1$; equivalently the depth complexity $D(f \circ g)$ of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded:** 2026-04-28 01:39 UTC
- **Entry ID:** 8047f7f8eea1

Statement

For all boolean functions $f: \{0,1\}^n \rightarrow \{0,1\}$ and $g: \{0,1\}^m \rightarrow \{0,1\}$ for which $HX^1(X_f)$ and $HX^1(X_g)$ admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace pairing on the KW-metric spaces obeys the sub-additive bound $\kappa(f \circ g) \geq \kappa(f) + \kappa(g) - C$, where C is a universal constant independent of n, m , and where $(X_{\{f \circ g\}}, d_{\{f \circ g\}})$ is built by the coarse product operation. In particular, witness cocycles $c_f \boxtimes c_g \in HX^1(X_{\{f \circ g\}})$ and tensor Roe operators $T_f \otimes T_g \in C_u^*[X_{\{f \circ g\}}]$ realize $\log|(c_f \boxtimes c_g, T_f \otimes T_g)| / \log(\text{propagation}(T_f \otimes T_g)) \geq \kappa(f) + \kappa(g) - C$.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 ($\kappa \leq \text{depth}$) to recover the KRW conjecture $D(f \circ g) \geq D(f) + D(g) - O(1)$. The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: $\langle c_f \boxtimes c_g, T_f \otimes T_g \rangle = \langle c_f, T_f \rangle \cdot \langle c_g, T_g \rangle$ (Connes-style product), while the propagation of $T_f \otimes T_g$ equals $\text{propagation}(T_f) \cdot \text{propagation}(T_g)$ under the depth-additive metric of coarse product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has $\kappa = O(1)$, so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composition
    X_f, d_f = build_X_d(f, k_f)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
    ~~~~~
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` ('int' object is not iterable) at `build_X_d` before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix `build_X_d` to enumerate vertices correctly (e.g., `X = [tuple(b) for b in itertools.product([0,1], repeat=n)]`) and rerun the 25-pair $\times$ 5-seed protocol.

Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity $\times$ Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. $AC^0[\oplus]$ regime, expressed via the coarse 1-cohomology $HX^1(X_f)$ and the Roe-trace pairing on $C_u^*[X_f]$
- **Recorded:** 2026-04-28 03:39 UTC
- **Entry ID:** 264c40c0d040

Statement

Let $f_n = \oplus\text{-tree}_n$ be the balanced binary tree of XOR gates on $n = 2^k$ inputs, with KW-metric space $(X_{\{f_n\}}, d_{\{f_n\}})$. Define the coboundary $\delta : C^0_{\text{coarse}}(X_{\{f_n\}}) \rightarrow C^1_{\text{coarse}}(X_{\{f_n\}})$ and the propagation-R Roe subalgebra $C_u^*[X_{\{f_n\}}]_R$. Then there exists a universal constant $\alpha \in (0,1)$ such that, for all $R \leq \alpha \log_2 n$, every coarse 1-cocycle c with propagation $\leq R$ is a coboundary (so its class in HX^1 is trivial), while for $R \geq k = \log_2 n$ there exist nontrivial classes c whose Roe-trace pairing satisfies $\log|c, T| / \log(\text{propagation}(T)) \geq k - O(1)$. Consequently $\kappa(f_n) \geq \log_2 n - O(1)$, matching $\text{depth}(f_n) = \Theta(\log n)$ up to constants and certifying axiom A1 sharply for an explicit recursive function.

Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known ($\text{depth}(\oplus\text{-tree}_n) = \log_2 n$).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation $R \approx \log_2 n$, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that k is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of ℓ_p uniform Roe algebras

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
           ~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on $R^*(n)$, $\dim HX^1_R$, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across $n \in \{4, 8, 16, 32\}$.